SIMATS ENGINEERING
C02-ASSESSMENT TOOL 1

COURSE NAME: Machine Learning COURSE CODE: ITA0611

Code of Conduct:

I _K.ASWINI(192312397)_ _ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

CO1: 1, A simple perceptron model is used to classify whether a student passes or fails based on two input features, namely study hours and attendance. The weights associated with these inputs are 0.6 and 0.4 respectively, and the bias value is –0.5. For a given student with study hours equal to 2 and attendance equal to 1, compute the net input to the perceptron and apply the step activation function to determine the final output. Based on the result, conclude whether the student passes or fails.

(5 Marks)

2. In a neural network, a single neuron is trained using the backpropagation learning rule. The initial weight is 0.5 and the learning rate is 0.1. For an input value of 2, the actual output produced by the neuron is 0.6, whereas the target output is 1. Calculate the error in the output and determine the weight update using the delta rule. Hence, compute the new updated weight after one iteration of training.

(5 Marks)

1. Perceptron Model (Pass / Fail)
Given:

- Weights:
  $w_1 = 0.6$, $w_2 = 0.4$
- Bias: $b = -0.5$
- Inputs:
  Study hours $x_1 = 2$
  Attendance $x_2 = 1$

Step 1: Compute Net Input

$$\text{Net Input} = (w_1 \cdot x_1) + (w_2 \cdot x_2) + b = (0.6 \times 2) + (0.4 \times 1) - 0.5 = 1.2 + 0.4 - 0.5 = 1.1$$

Step 2: Apply Step Activation Function
Step function:

- If net input ≥ 0 → Output = 1 (Pass)
- If net input < 0 → Output = 0 (Fail)

Since:

$$1.1 \geq 0$$

Output = 1

Final Conclusion:
The student **passes**

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2. Backpropagation (Delta Rule)
Given:

- **Initial weight $\boldsymbol{w = 0.5}$**
- **Learning rate $\boldsymbol{\eta = 0.1}$**
- **Input $\boldsymbol{x = 2}$**
- **Actual output $\boldsymbol{y = 0.6}$**
- **Target output $\boldsymbol{t = 1}$**

Step 1: Calculate Error

$$\mathbf{Error} = \boldsymbol{t - y = 1 - 0.6 = 0.4}$$

Step 2: Apply Delta Rule

$$\boldsymbol{\Delta w = \eta \cdot (t - y) \cdot x = 0.1 \times 0.4 \times 2 = 0.08}$$

Step 3: Update Weight

$$\boldsymbol{w_{\text{new}} = w_{\text{old}} + \Delta w = 0.5 + 0.08 = 0.58}$$

Final Answers:

- **Error = 0.4**
- **Weight update (Δw) = 0.08**
- **New weight = 0.58**